DATA SCIENCE AND BUSINESS INFORMATICS

VISUAL ANALYTICS

Analyzing Bias in COOTEFOO: Solving VAST Challenge 2025

UNIVERSITÀ DI PISA

Alice Calderini

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Contents

1 Introduction

This project aims to provide a solution to **Mini Challenge 2** of the **VAST Challenge 2025**. The challenge concerns Oceanus, whose economy has historically been based on fishing and which, more recently, has seen a growing presence of the tourism sector. The local government has established a commission, COOTEFOO, with the task of monitoring the island's economy.

Fishing is Living and Heritage (**FILAH**) accuses the commission of having a bias in favor of tourism, while Tourism Raises OceanUs Together (**TROUT**) argues that the bias is instead in favor of fishing. The **journalist** Moray acquired the data provided by FILAH and TROUT, as well as additional data not collected by the two organizations. The objective of the project is to build an interface that supports the journalist in analyzing the data, allowing her to verify the validity of the criticisms directed at the commission.

The visual analytics interface was implemented as a Single Page Application (SPA) using **Vue.js** as the frontend framework, **TailwindCSS** for styling, and **D3.js** for data-driven visualization and rendering, including its `d3-geo` module for geographic projections.

2 Data Understanding and Data Preparation

This section summarizes the *data understanding* process conducted on the three provided knowledge graphs (FILAH, TROUT, journalist) and the consequent *data preparation* choices, aimed at building the tables used in the visual analysis interface. The knowledge graphs contain: FILAH 396 nodes and 765 edges; TROUT 164 nodes and 378 edges; journalist: 740 nodes and 2436 edges.

These are complemented by two auxiliary geographic sources: `road_map.json`, a graph consisting of 2664 nodes and 3420 edges, in which each node represents a point with `latitude`/`longitude` coordinates and `zone`; and `oceanus_map.geojson`, which provides the geographic boundaries of 10 islands/nature reserves and the coordinates of 7 inhabited centers and 12 navigation buoys.

2.1 Graph Schema and Node Types

The three knowledge graphs share the same conceptual schema: a directed multigraph in which each node exposes a `type` field and each edge a `role` field. Eight node types were identified:

`discussion`, `entity.organization`, `entity.person`, `meeting`, `place`, `plan`, `topic`, `trip`.

The edges connect the nodes through six distinct roles, identified by the `role` field:

- `part_of`: connects a `meeting` to a `discussion` or to a `plan`, indicating the meeting in which the discussion or plan was discussed or created;
- `about`: connects a `discussion` to the `plan` or `topic` it refers to;
- `participant`: connects a `discussion` or a `plan` to an entity (`entity.person` or `entity.organization`) and contains the attributes `sentiment`, `reason` and `industry`;
- `refers_to`: connects a `discussion` to a mentioned `place`;
- `plan`: connects a `plan` to the `topic` it belongs to;
- `travel`: connects a `plan` to a `place`;
- the connections `trip`→`entity.person` and `trip`→`place` do not have an explicit role; they are used to represent the physical movements of board members.

A subset of nodes did not report any `type` field and is therefore not included in Table 2.1. These include `10803677425` (and other nodes attributable to the `place` type), `Bay Harvest Corporation`, `Harbor Odyssey Tours`, `Sean`, `concert_Travel_Harborfront_Market` and `name_harbor_area_Meeting_`

Node type	journalist	FILAH	TROUT
entity.person	6	3	6
entity.organization	8	8	8
meeting	16	12	13
discussion	101	58	39
plan	74	41	33
topic	15	14	14
place	159	59	30
trip	342	189	18

Table 2.1: Number of nodes by type in the three knowledge graphs.

`11_Harbor_Odyssey_Tours`. It was verified that, with the exception of nodes of type `place`, all the others appear exclusively in journalist and are absent from FILAH and TROUT.

Correction of missing types. The type of nodes lacking the `type` field was inferred based on the edges connected to them:

- a node reached by a `refers_to` or `travel` edge was classified as a node of type `place`;
- a node reached by an incoming `about` edge originating from a `discussion` was classified as a node of type `plan`;
- a node reached by a `participant` edge was classified as `entity.person` or `entity.organization`. The distinction was made manually: `Sean` was classified as a person based on the name, while `Bay Harvest Corporation` and `Harbor Odyssey Tours` were classified as organizations.

The `place` with `id` 10803677425 ("Harbor Odyssey Tours") is present in journalist as a node without any attribute other than the `id`: all its attributes (lat, lon, zone, city, name) were retrieved from `road_map`.

2.2 Consistency Between the Sub-Datasets and the Complete Dataset

Missing values. Missing values are concentrated on `participant` edges, in which `sentiment`, `reason` and `industry` are simultaneously absent in 9 FILAH edges, 12 TROUT edges and 19 journalist edges. The `place_name` field is frequently null, which is plausible since not all `place` nodes are necessarily associated with a name.

When an edge is present in both a sub-dataset and journalist, the set of missing fields is identical in the two datasets. **The sub-datasets therefore do not hide additional information compared to journalist.**

Coverage by topic and by meeting. FILAH does not contain any initiative related to the topic `concert`, while TROUT does not contain any initiative related to `seafood_festival`. The absence is total: if a topic is missing from a sub-dataset, all the `discussion` and `plan` nodes that are connected to that topic in journalist are also missing. Therefore, there are no inconsistencies, such as initiatives lacking their corresponding topic.

The same pattern was found for meetings: FILAH does not cover `Meeting_13`–`Meeting_16`, while TROUT does not cover `Meeting_13`–`Meeting_15`. For each meeting absent from a sub-dataset, all the `discussion` and `plan` nodes connected to it in journalist are also absent.

For meetings that are present, however, not all associated initiatives are necessarily reported. In most cases, only some participants are missing; in two cases, in FILAH, the entire `plan` connected to the `discussion` is missing.

Overall, **the sub-datasets do not constitute partial fragments of the information: an item of information is either present or absent, but it is not reported in a partially incomplete form.**

2.3 Meeting–Discussion–Plan Structure

A `meeting` is connected to a `discussion` (and every `discussion` is always associated with a meeting) through a `part_of` edge. The `discussion` is connected to a `plan` through an `about` edge, while the `meeting` is also directly connected to the `plan` through an additional `part_of` edge. It represents the meeting in which the plan was created.

The following cardinality properties were verified: each `trip` is associated with a single person, who may nevertheless visit multiple places (up to 12 recorded stops), each with its own time; each `discussion`/`plan` is associated with a single `topic` and a single `place`; each `discussion` is connected to a single `plan`, while a `plan` can be discussed in multiple `discussion` nodes, corresponding to its temporal checkpoints.

When the connected `discussion` and `plan` both report a `place`, they coincide in all cases except one. The completion discussion of the plan `waterfront_market_Travel_Harborfront_Market` in fact refers to *Paakland Elementary*, while the plan refers to *Harborfront Market*.

The relationship `discussion` $\subseteq$ `plan` does not hold, in general, with respect to participants: there are people present in the discussion and not in the plan, and vice versa. However, when the same person appears in both, their `sentiment` is always identical. Similarly, when the same person appears in multiple checkpoints (discussions) of the same plan, the sentiment remains constant over time: no case of a change in opinion between one checkpoint and another was found.

A person's sentiment also remains unchanged across different plans that share the same topic. Finally, it was verified that, when an entity (person or organization) expresses an opinion on a topic, the corresponding `sentiment` value is identical in all datasets in which the entity appears. **FILAH and TROUT can therefore omit some opinions, but cannot alter their content.** These observations guided the construction of the final tables (Sect. 2.4) and of some graphs in the visual analytics interface (Sect. 3.4.1 and Sect. 3.4.3).

It was also verified whether `industry`, an attribute of the `participant` edge, constituted a unique superclass of `topic`. Out of 15 topics, 14 present a constant set of industries for all participants. The only exception is `fish_vacuum`, associated with both *small vessel* and *large vessel* depending on the participant. This is not a disagreement, but rather two different sector perspectives on the same technology: a "fish vacuum" is relevant to both small-scale and large-scale fishing.

2.4 From Findings to the Final Tables

The properties verified in the previous phase directly guided the structure of the final tables used in the analysis.

Merging Discussion and Plan into a single `Initiative` entity. Since connected `discussion` and `plan` nodes always share the same topic and place, except for the case described above, and since a person's sentiment does not change between discussion and plan or between successive checkpoints, the two entities were merged into a single `INITIATIVES` table. The conflict between *Paakland Elementary* and *Harborfront Market* was reported in the table with the attribute `place_id_conflict`.

The initiative identifier corresponds to the id of the `plan` when present, or to that of the `discussion` when the latter has no connected plan. This also eliminates the redundancy of the sentiment, which remains the same for the same initiative. This is instead stored in the `INITIATIVE_PARTICIPANTS` table, which contains the participants (people or organizations) in the initiative, together with their corresponding `sentiment`, `reason` and `industry`.

Temporal evolution. The progression of an initiative through successive meetings (from `introduced` to `completed`) is contained in a separate table, `INITIATIVE_STATUS_TIMELINE`, with one row for each checkpoint of the initiative and the meeting in which that specific checkpoint occurred.

Provenance flags `in_filah`/`in_trout`. Since it was verified that journalist constitutes a strict information superset, the three datasets were collapsed into a single version of the tables, introducing two boolean columns to track provenance rather than maintaining three separate copies.

In the `INITIATIVES` table, the flag is calculated according to an OR logic: the initiative is considered present if the `plan` is present or at least one of the `discussion` is present. The provenance flag was also replicated in the `INITIATIVE_PARTICIPANTS` table, in which each participation (person $\times$ initiative) reports whether it is known in FILAH and/or TROUT.

Places. The `lat` field contains values between -165.96 and -164.33: a range incompatible with a valid latitude (which must fall within $[-90, 90]$), suggesting a swap with longitude. The suspicion was confirmed by cross-referencing the `id` of the `place` nodes with `road_map`. The coordinates were therefore corrected in the new `places.json` file. The `city_name` field was also added, retrieved from `road_map`, as it was absent from journalist.

When both are present, the `name` and `label` fields of the `place` nodes in journalist coincide. In the final tables, a single `name` attribute was therefore retained; when `name` is null but `label` is present, the latter was used as the name of the place. This also allowed us to recover some `place` names with null names.

Final tables. The data preparation produces ten final tables, summarized in Table 2.2.

Table	**Main attributes**
PERSONS	id, name, role, role_note, in_filah, in_trout
ORGANIZATIONS	id, in_filah, in_trout
TOPICS	id, short_topic, long_topic, in_filah, in_trout
MEETINGS	id, date_label, in_filah, in_trout
INITIATIVES	id, topic_id, place_id, short_title, long_title, plan_type, created_meeting_id, final_status, n_checkpoint, in_filah, in_trout
INITIATIVE_PARTICIPANTS	initiative_id, entity_id, entity_type, sentiment, reason, industry, in_filah, in_trout
INITIATIVE_STATUS_TIMELINE	initiative_id, meeting_id, status
PLACES	id, name, city_name, latitude, longitude, zone, zone_detail, in_filah, in_trout
TRIPS	id, person_id, date, start, end, in_filah, in_trout
TRIP_STOPS	trip_id, place_id, time

Table 2.2: Schema of the final tables produced by the data preparation.

3 Design and Functioning of the Visual Analytics Interface

3.1 State of the Art

To define the choices for sentiment representation, Kucher, Paradis and Kerren [1], *The State of the Art in Sentiment Visualization* (Computer Graphics Forum, 2018), were consulted. This survey proposes a categorization of the main sentiment visualization techniques. The work confirms the validity of some of the choices adopted in this project, in particular the use of color to encode opinion polarity and the representation of sentiment through an entity×topic matrix, corresponding to the heatmap discussed in Section 3.4.1. The system therefore builds on established techniques, favoring simple representations over more experimental approaches designed for larger datasets.

3.2 Model and Communication Pattern

The interface was named *Inspecting Currents*, a name that captures the three main pillars of the project. On the one hand, the *investigative* register. On the other, the *maritime* reference to the setting of Oceanus, of which "currents" is a direct metaphor. Finally, "currents" also refers to *currents of thought*, the narratives, often in conflict with one another, through which the three datasets (journalist, FILAH, TROUT) describe the COOTEFOO committee's positions on the same initiatives. "Inspecting currents" therefore means "inspecting the currents of opinion" running through the committee.

The interface adopts a multi-page architecture organized into three views: *COOTEFOO Members*, *Map & Movements*, and *Weigh-in*, each corresponding to a phase of the bias investigation. The first allows users to explore, person by person and topic by topic, the sentiment expressed by COOTEFOO committee members and organizations; the second provides behavioral context by showing where and how frequently members move; the third enables a quantitative comparison between industries and datasets (journalist, FILAH, and TROUT), directly measuring the extent of the bias.

The interface is *reader-driven*: it does not present predefined filters or mandatory paths, but the structure of the three pages implicitly guides an analytical path from individual detail to behavioral context and finally to quantitative aggregation.

A common navbar allows users to move from one view to another at any time, while a shared sidebar exposes the controls that act on the widgets of the current page: dataset selection, selection of a person across all widgets and, where applicable, aggregation of the *small vessel*/*large vessel* categories into *fishing*. Both person selection and the *fishing* aggregation behave as toggles: clicking on an already selected element cancels the selection.

3.3 Design Choices: Colors, Shapes, and Interactions

Colors. The base palette recalls the maritime theme of Oceanus and the values of clarity and reliability that the project aims to communicate to a journalistic audience, through a range of blues. Color nevertheless also plays a precise semantic role:

- a very dark blue (■ `#0F172A`) is reserved for the navbar and for the controls that reconfigure the data shown in the page widgets, such as dataset selection and the aggregation of *small vessel*/*large vessel* into *fishing*;
- a lighter blue (■ `#185ead`) identifies the *selected person*, maintaining the same color encoding wherever that person appears in the interface;
- purple (■ `#6d48b5`) identifies the selection of a topic or industrial category. It was chosen outside the blue range to distinguish it from the green and yellow tones used for sentiment;
- in bar charts comparing discrete categories that cannot be ordered relative to one another, such as *Stops by zone* and *Initiatives by industries and dataset*, more heterogeneous colors are used. A single color scale would in fact implicitly suggest an ordering or relationship between categories that does not exist in the data;
- for the representation of sentiment, through heatmaps or bars, a red-yellow-green diverging scale is used: green indicates positive sentiment, yellow intermediate values, and red negative sentiment.

Shapes. The visual vocabulary was limited to simple shapes, mainly squares and rectangles, avoiding unnecessary geometries. The heatmap cells, bar segments, and widget containers are therefore rectangular. The widgets are arranged according to a regular grid: vertically when independent, horizontally when complementary.

Interactions. All widgets provide tooltips on mouse hover (*hover*), showing detailed information without overloading the overall view. A *click* (or, in some cases, a *hover*) also allows users to select an item (person, zone, topic, or trip), updating its highlighting in the other widgets on the same page (*linked views*). These interactions are complemented by data reconfiguration controls, such as dataset selection and the aggregation of *fishing*.

3.4 Interface Structure: Pages and Interactions

3.4.1 COOTEFOO Members

The main widget is a heatmap with people and organizations on the vertical axis and topics on the horizontal axis. The cells represent the **sentiment of the entity toward the topic**. As discussed in Section 2.3, for the same person, the sentiment associated with a topic remains unchanged across different discussions; **it is therefore represented only once, regardless of how many times the opinion is expressed.**

The toggle *By topic / By industry* allows users to replace topics with industries. In *By topic* mode, it is also possible to filter the columns by industry: when some topics are hidden, the remaining cells automatically expand to occupy the available space. An additional filter allows users to display people, organizations, or both. Selecting a person from the sidebar or from any widget updates their highlighting throughout the page; the same mechanism is available for organizations and topics, which can be selected by clicking on their name. In the heatmap, selecting an entity also fades the cells belonging to the other rows.

Next to the heatmap, the *Heatmap insights* panel summarizes, for the active dataset, the number of people and organizations with known opinions and the number of covered topics. When the dataset is changed, it also highlights which people or topics are absent or lack an opinion compared with the complete picture provided by journalist. The Figure 3.1 shows the heatmap with examples of active filters in the sidebar and within the widget.

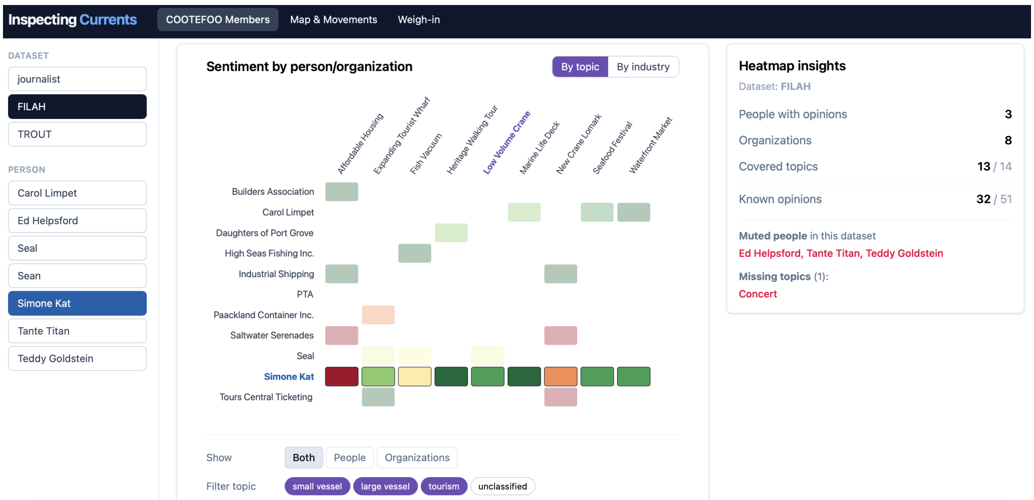

Figure 3.1: Heatmap

The *Average sentiment by topic* widget (Fig. 3.2) presents an ordered horizontal bar chart, colored according to the same diverging scale used in the heatmap. Each bar represents the **average sentiment for a topic**. Selecting a topic from any widget causes the *Who says that?* panel to list the individual opinions collected, together with their textual rationale. A *Reset* button, visible only when a topic is selected, allows the selection to be cleared.

When a topic is selected, the other bars are instead faded, highlighting the chosen topic. Selecting a person overlays a white circle on the bars for each topic on which they expressed an opinion, positioned at their *own* sentiment value for that topic. **Comparing the position of the circle with the length/direction of the bar makes it possible to see at a glance whether that person aligns with the committee's average position on a topic, or is instead an outlier.**

Finally, the *Activities of selected entity* widget shows, for the selected entity (person or organization),

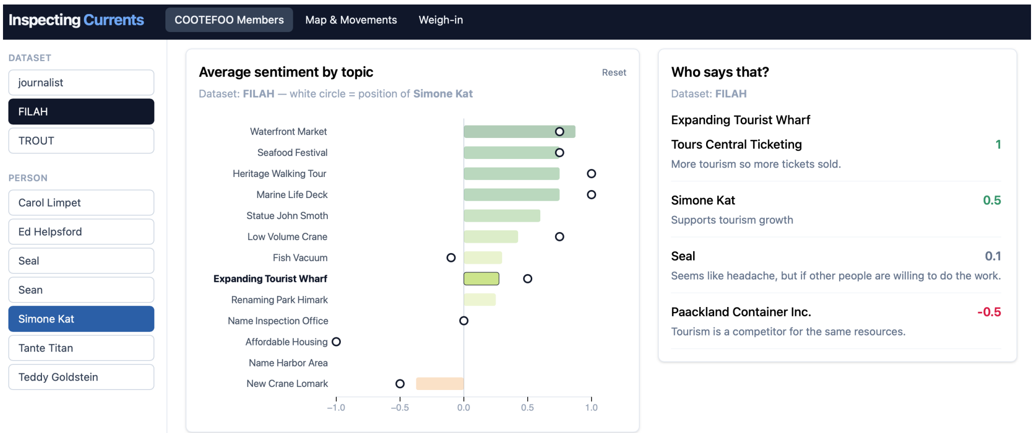

Figure 3.2: Average sentiment by topic

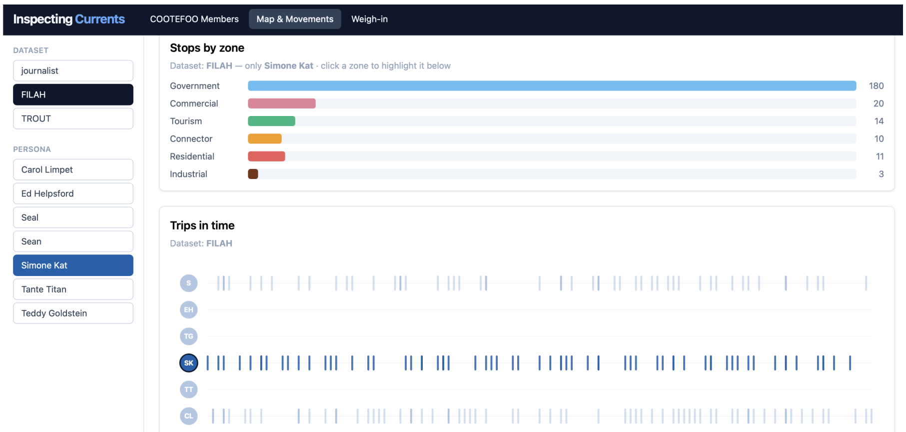

Figure 3.3: Members Trips

the number of initiatives, topics, meetings, and trips on which they expressed an opinion, grouped by industry; it also provides the textual details of the initiatives they participated in. Organizations can be selected (and deselected) from the other widgets in which they appear, while for people the sidebar also provides a shortcut.

3.4.2 Map and Movements

The *Steps by zone* widget (Fig. 3.3) shows the number of stops, rather than distinct trips, made in each type of zone (*government*, *commercial*, *tourism*, etc.). Selecting a zone highlights the trips that include it in the widget below.

Trips in time (Fig. 3.3) represents each trip as a vertical bar positioned along the temporal axis, with one row per person on the vertical axis. Selecting a person's initial highlights their trips and updates the *Steps by zone* bar chart, showing the number of stops made by that person in each zone. Hovering over the initial provides the same information in textual form, allowing for faster scanning, while clicking is preferable for comparing proportions in the bar chart.

Hovering over an individual trip shows the stops visited, each identified by a colored dot according to the type of zone, using the same encoding as *Stops by zone*. Selecting the trip instead draws its route on the map below.

The map (Fig. 3.4) shows the places known in the active dataset as colored points according to their zone, overlaid on the outlines of the archipelago's islands. It is possible to zoom and navigate by dragging; hovering over each point or island displays the corresponding information. It should be emphasized that the points shown are those present in `places.json`, i.e., the `place` entities actually cited in the original knowledge graphs, rather than the entire road network available in `road_map.json`:

this is a simplification choice, given that the remaining points in `road_map` have no relationship with the initiatives or movements of the committee members, and including them would clutter the map without adding analytical value.

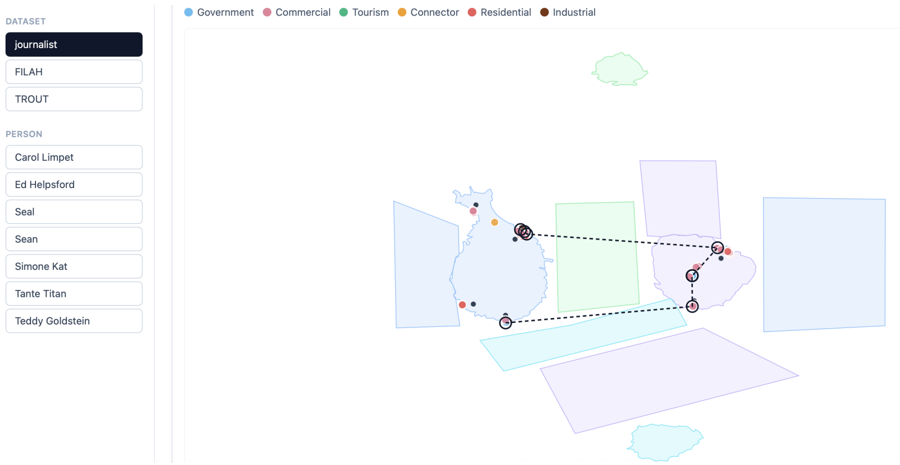

Figure 3.4: Oceanus Map

3.4.3 Weigh-in

The *Weighing the bias among industries* widget (Fig. 3.5) allows users to select **two groups of industries to compare** (Group A and Group B) among *tourism*, *small vessel*, *large vessel*, and *unclassified*, the latter corresponding to topics with no associated industry.

For each person or organization that expressed an opinion on at least one topic belonging to the selected industry, the sentiment associated with the relevant topics is summed; the group total is shown at the top, in green if positive and in red if negative. As discussed in Section 3.4.1, the sentiment of a topic is counted only once, regardless of the number of initiatives in which it is expressed.

Clicking on an industry already present in the same group replaces the current selection, with the exception of the *small vessel*/*large vessel* pair, which is aggregated within the same group because, in the context of the analysis, both can be traced back to fishing. The *Aggregate fishing* button in the sidebar provides an equivalent shortcut, directly merging the two categories into a single *fishing* category, both in this widget and in the following one.

Selecting a person or organization in either of the two lists causes the side panel to display the details of their opinions contributing to the corresponding total for that industry.

The *Initiatives by industry and dataset* widget completes the analysis by considering the dimension of *coverage*, namely the number of known initiatives in each dataset for each industry. **One possible manifestation of bias may consist in a dataset reporting fewer initiatives related to a given industry.**

3.5 Use Case: Analyzing Bias Between Tourism and Fishing

The user, who may be the journalist, starts from the *COOTEFOO Members* page. By filtering by industry and analyzing the heatmap, they observe that Simone Kat has a negative opinion of large vessel, while Teddy Goldstein has a negative opinion of tourism. Changing the dataset from the sidebar, they notice that in FILAH Teddy Goldstein's opinion disappears completely, while in TROUT Simone Kat's opinion is missing. They also notice that one person, Sean, appears only in journalist and has very little data: one initiative, for which sentiment is not recorded.

In the *Heatmap insights* panel, they also verify that, in the two datasets, three people are "mute", meaning that they are either absent or have no recorded opinions. In FILAH, Ed Helpsford, Tante

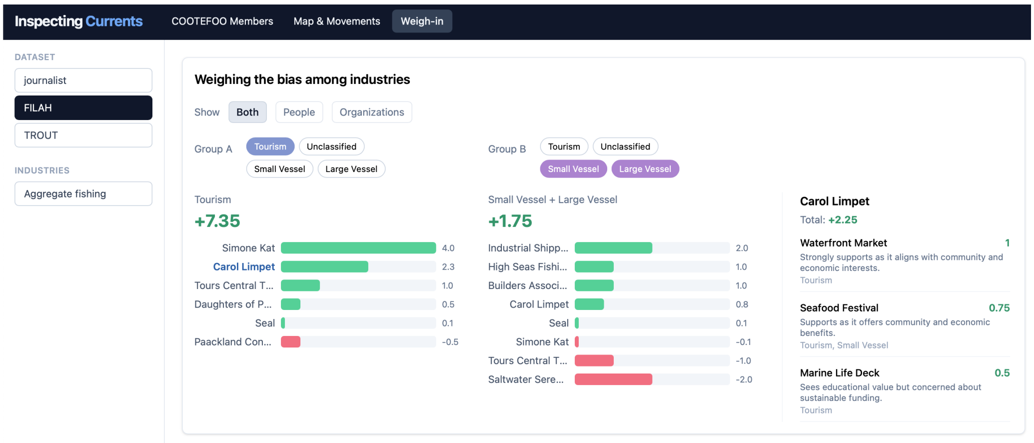

Figure 3.5: Weighing the bias among industries

Titan, and Teddy Goldstein are absent; in TROUT, all are present, but the sentiment of Carol Limpet, Simone Kat, and Tante Titan is not reported.

In the *Average sentiment by topic* widget, the journalist dataset shows the highest sentiment values for topics related to tourism, such as *waterfront market* and *heritage walking tour*, while *new crane lomark*, related to fishing, has negative sentiment. In FILAH, the pattern is repeated, with *marine life deck* (tourism) showing an even higher sentiment. In TROUT, on the other hand, the topics with higher sentiment compared with journalist include *affordable housing* and *fish vacuum*, both related to fishing. This provides an initial indication of a **systematically more favorable representation of fishing in TROUT**.

Before the quantitative comparison, the journalist also checks coverage in terms of movements by moving to the *Map & Movements* page. In FILAH, the people previously identified as absent (Ed Helpsford, Tante Titan, and Teddy Goldstein) are indeed completely absent from the movement data. In TROUT, on the other hand, almost all members remain represented (only Ed Helpsford is missing), but with a drastically reduced number of trips, providing a further indication of possible **sampling bias**.

They then move to the *Weigh-in* page, comparing tourism and *small vessel + large vessel*, aggregated into *fishing*. In the journalist dataset, the ratio is **10.35 versus 7.00**, indicating a limited difference. In FILAH, the ratio becomes **7.35 versus 1.75**, with a much *larger gap in favor of tourism*. In TROUT, the pattern is reversed: **1.6 versus 5.6**, highlighting a **clear bias in favor of fishing** and quantitatively confirming the suspicion that emerged from the topic analysis.

Filtering to people only, the pattern remains identical across the three datasets; considering organizations only, the ratio is always 1 to 1. The observed bias is therefore determined by the individual opinions of committee members, rather than by those of the organizations involved.

Finally, in the *Initiatives by industries and dataset* widget, TROUT reports far fewer initiatives for *tourism* than FILAH, while the proportion in FILAH is closer to that of journalist. This provides further confirmation, this time in terms of coverage rather than tone, that **TROUT tends to marginalize the tourism component in favor of fishing-related initiatives**.

3.6 Answers to the VAST Challenge Questions

Is there evidence of bias in the COOTEFOO member actions in either dataset? We interpret "actions" as the positions taken by committee members on committee initiatives, expressed through sentiment. Yes, there is evidence of bias in these positions: in FILAH, the aggregated sentiment leans clearly toward tourism, while in TROUT it leans clearly toward fishing.

Is the committee as a whole biased? Provide visual evidence. As shown in Section 3.5:

- **journalist**: +10.35 (*tourism*) versus +7.00 (*fishing*), indicating a slight imbalance toward tourism, with both industries evaluated positively overall;
- **FILAH**: +7.35 versus +1.75, with a significantly larger gap in favor of tourism;
- **TROUT**: +1.6 versus +5.6, with a complete reversal of the pattern and a **clear bias in favor of fishing**.

Considering the most complete dataset, journalist, the committee is therefore **slightly biased toward tourism**, but not to an extreme extent. The two biased datasets instead accentuate the imbalance in opposite directions.

Are the accusations of TROUT strengthened, weakened or unchanged when taken in context of the whole dataset? Weakened. The journalist dataset shows a committee with a limited bias toward tourism, far from the picture described by TROUT, which instead shows an imbalance in favor of fishing.

Pick at least one COOTEFOO member accused by TROUT. Illustrate how your understanding of their activities changed when using the more complete dataset. Teddy Goldstein, one of the members presented by TROUT as opposed to tourism and favorable toward fishing, maintains the same sentiment between journalist and TROUT, as discussed in Section 2.3: tourism −1.00, fishing +2.50.

What changes is the context in which his position is interpreted. In journalist, his strongly negative opinion of tourism is counterbalanced by at least five other members with clearly favorable positions toward tourism, including Simone Kat (+4.00), Tante Titan (+2.50), and Carol Limpet (+2.25). In TROUT, three of these counterbalancing positions are completely missing, making Teddy Goldstein's position appear representative of the committee, whereas in the complete picture it is a **minority position**.

What are the key pieces of evidence missing from the original TROUT data that most influenced the change in judgement? As already described, the main element is the absence of the tourism-favorable opinions of Simone Kat, Tante Titan, and Carol Limpet, which contributes to shifting the overall balance in TROUT toward fishing.

Whose behaviors are most impacted by sampling bias when looking at the FILAH dataset in context of the other data? Ed Helpsford and Teddy Goldstein, both absent from FILAH. Ed Helpsford is moderately positive toward both industries, but more strongly toward fishing (tourism +1.50, fishing +2.00); Teddy Goldstein, on the other hand, is the only member with negative sentiment toward tourism (tourism −1.00, fishing +2.50). Their absence therefore changes the balance of the positions represented in the dataset.

Illustrate the bias of the FILAH data in the context of the whole dataset. FILAH is biased toward tourism. Removing Ed Helpsford and Teddy Goldstein leaves members whose overall pro-tourism sentiment is higher than their pro-fishing sentiment, increasing the gap from +3.35 in the journalist dataset (10.35 versus 7.00) to +5.60 in FILAH (7.35 versus 1.75), almost doubling the difference.

Bibliography

[1] Kostiantyn Kucher, Carita Paradis, and Andreas Kerren. "The State of the Art in Sentiment Visualization". In: *Computer Graphics Forum* 37.1 (2018), pp. 71–96. DOI: 10.1111/cgf.13217.